北京交通大學 BEIJING JIAOTONG UNIVERSITY

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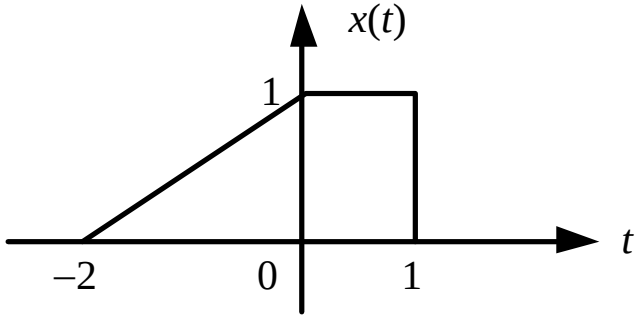
Basic operations of continuous-time signals

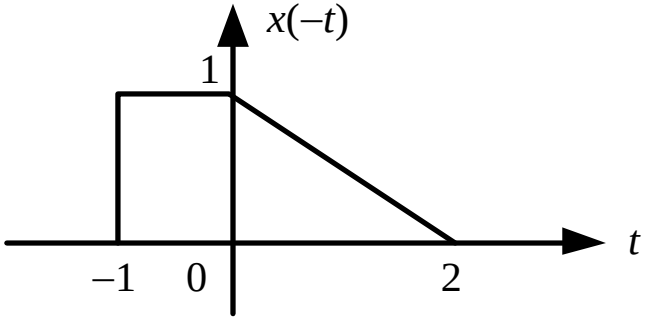
- signal reversal
- signal compression
- signal extension
- signal shifting
- signal addition
- signal multiplication
- signal differentiation
- signal integration

1. signal reversal

$x(t) \rightarrow x(-t)$

Flip $x(t)$ 180° around the vertical axis



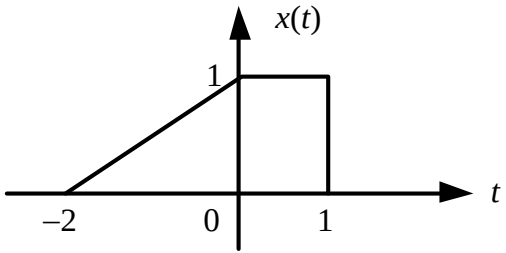


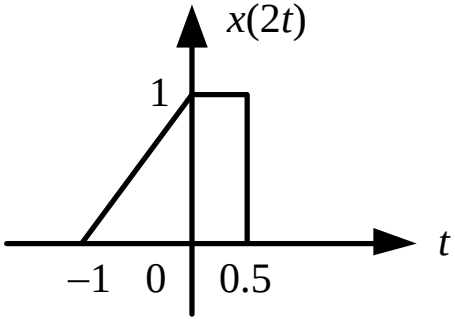
2. Compression and extension

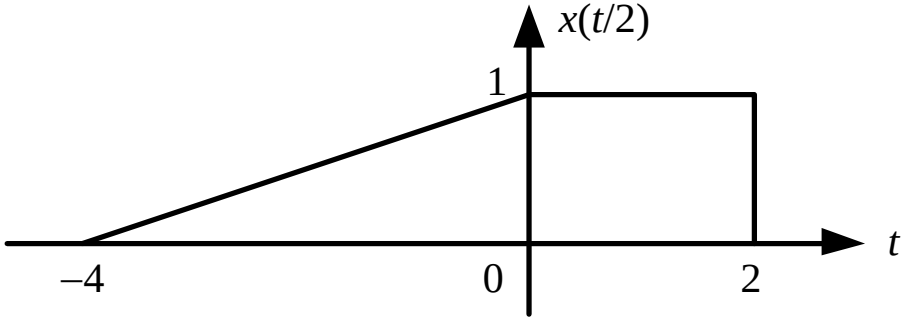
$x(t) \rightarrow x(at) \quad a>0$

If $0<a<1$, $x(at)$ is extension of $x(t)$

If $a>1$, $x(at)$ is the compression of $x(t)$





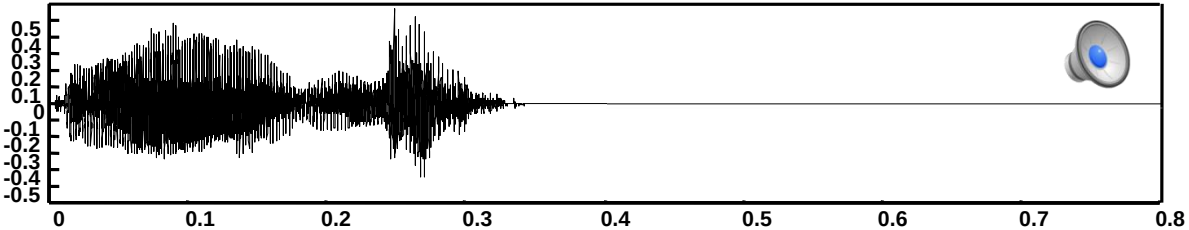


2. Compression and extension

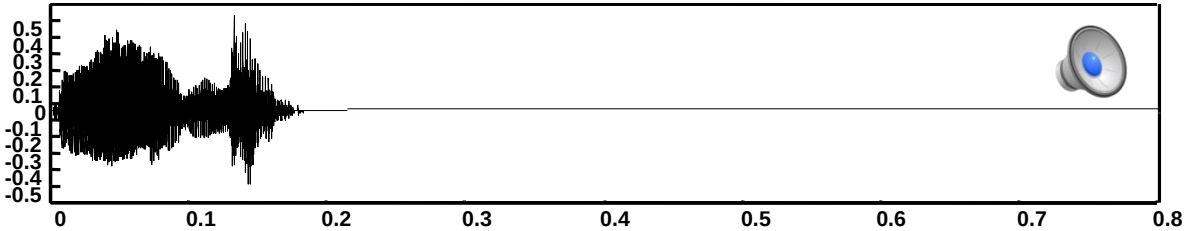
Fast forward/backward, slow forward/backward in the playback of video files

Normal playback $x(t)$

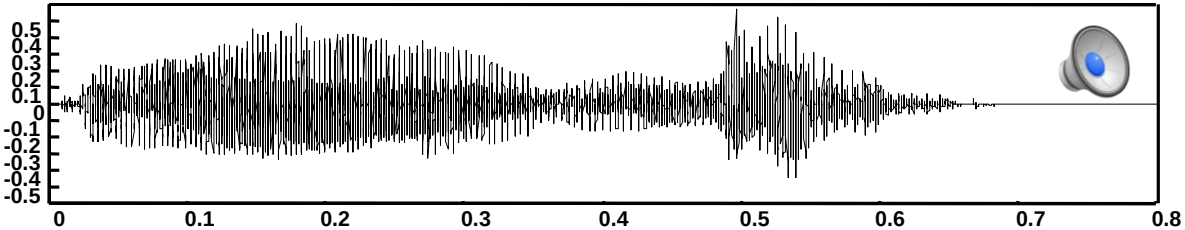
Voice signal (“对了”)
Sampling frequency = 22050Hz



Fast forward playback $x(2t)$



Slow forward playback $x(t/2)$

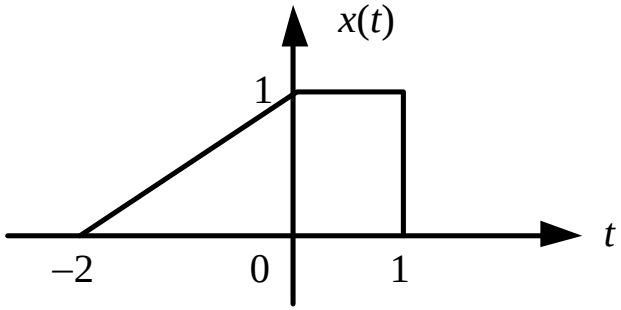


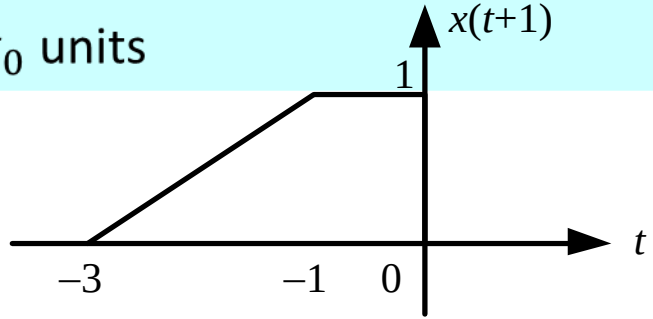
3. Signal shifting

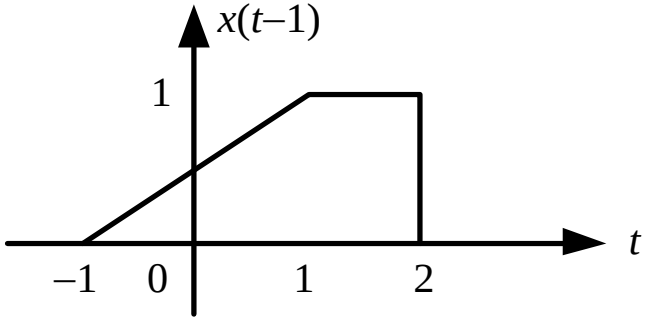
$x(t) \rightarrow x(t \pm t_0) \quad t_0 > 0$

$x(t-t_0)$ indicates that the signal moves right by t_0 units;

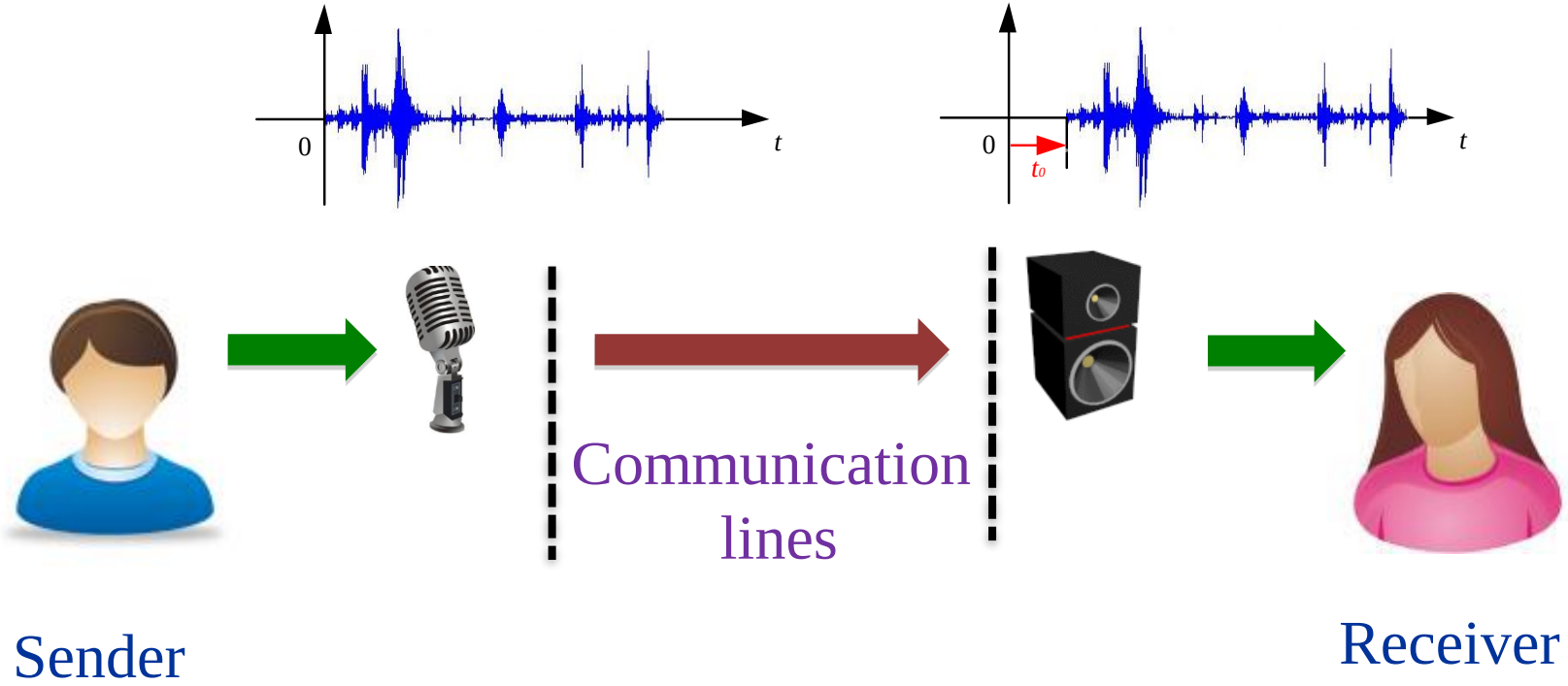
$x(t+t_0)$ indicates that the signal moves left by t_0 units





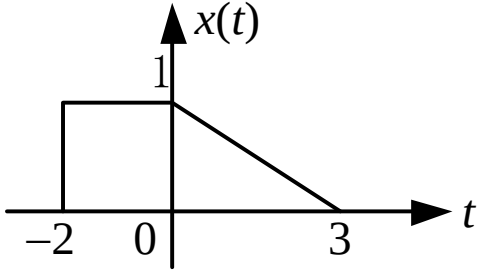


3. Signal shifting

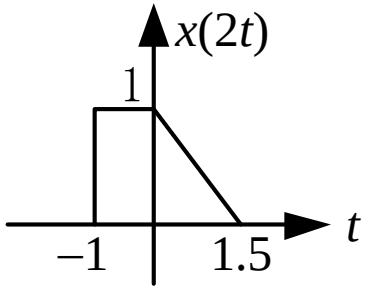


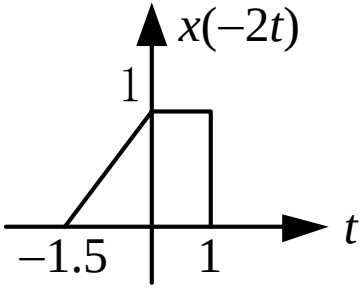
Remote communication causes signal delays

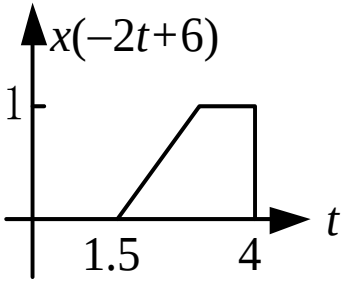
***Example*. The waveform of $x(t)$ is shown as below.**
Try to draw the waveform of $x(6-2t)$.



$$x(t) \xrightarrow[t\to 2t]{\text{Compression}} x(2t) \xrightarrow[t\to -t]{\text{Reversal}} x(-2t) \xrightarrow[t\to t-3]{\text{Right shifting}} x(-2(t-3))$$



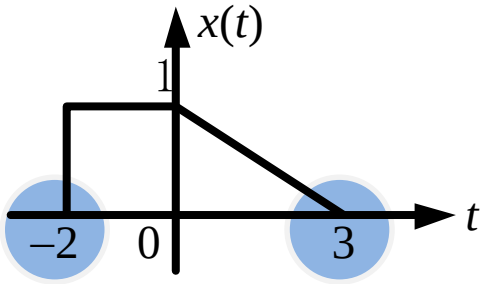




Endpoint function value invariance method

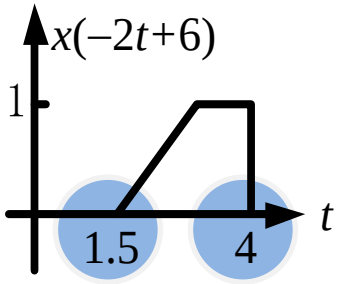
Basic idea: The flip, contraction and time-shift operation of the signal is only a linear transformation of the function's independent variables, and the function value of the signal endpoint before and after the transformation is unchanged.

Try to draw a waveform of $x(6\text{-}2t)$.



$$-2 = 6 - 2t_{11} \qquad t_{11} = 4$$

$$3 = 6 - 2t_{22} \qquad t_{22} = 1.5$$



Endpoint function value invariance method

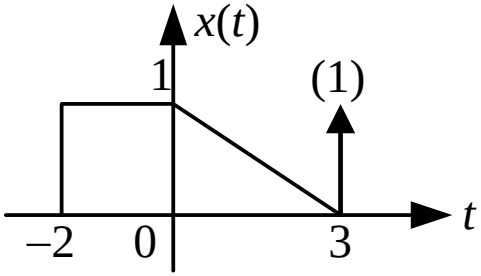
Basic idea: The flip, contraction and time-shift operation of the signal is only a linear transformation of the function's independent variables, and the function value of the signal endpoint before and after the transformation is unchanged.

$$x(t_1) = x(at_{11} + b) \qquad t_{11} = \frac{1}{a}(t_1 - b)$$

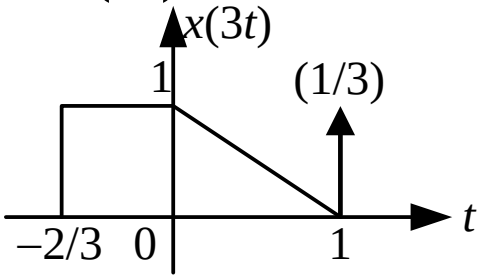
$$x(t_2) = x(at_{22} + b) \qquad t_{22} = \frac{1}{a}(t_2 - b)$$

Single choice 1point

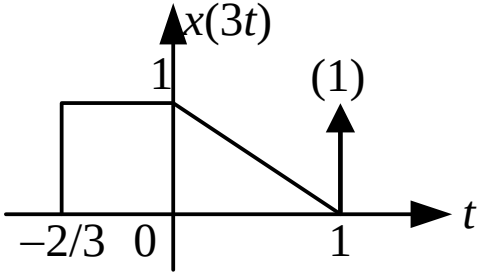
The waveform of the known signal $x(t)$ is shown in the following figure, and the waveform of $x(3t)$ is



A



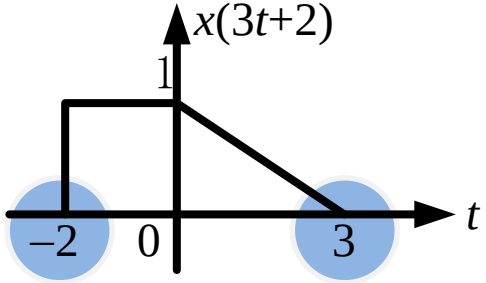
B



Endpoint function value invariance method

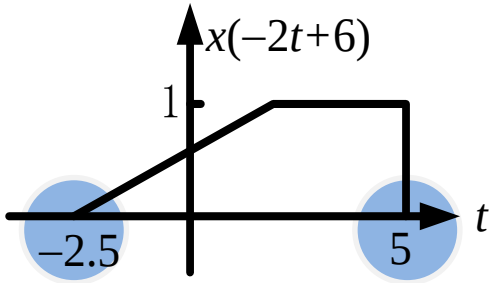
Basic idea: The flip, contraction and time-shift operation of the signal is only a linear transformation of the function's independent variables, and the function value of the signal endpoint before and after the transformation is unchanged.

Example: The known waveform of $x(3t+2)$ is shown in the figure, try to draw the waveform of $x(6-2t)$



$$3\times -2+2=6-2t_{11} \qquad t_{11}=5$$

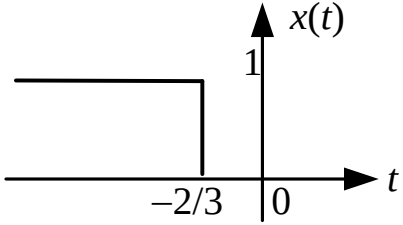
$$3\times 3+2=6-2t_{22} \qquad t_{22}=-2.5$$



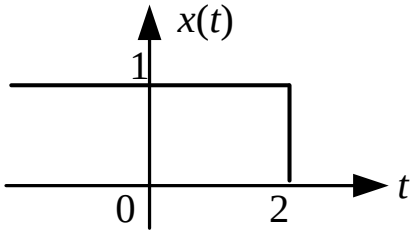
Single choice 1point

The waveform of *u(2-3t)* is

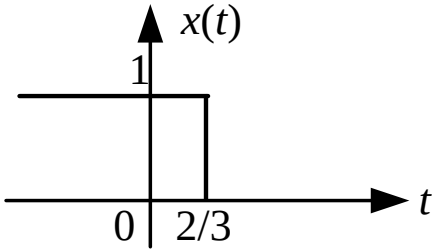
A



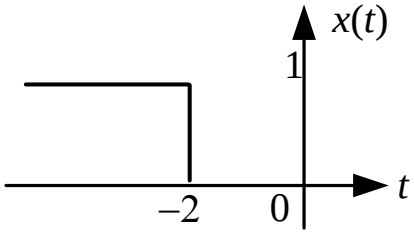
B



C

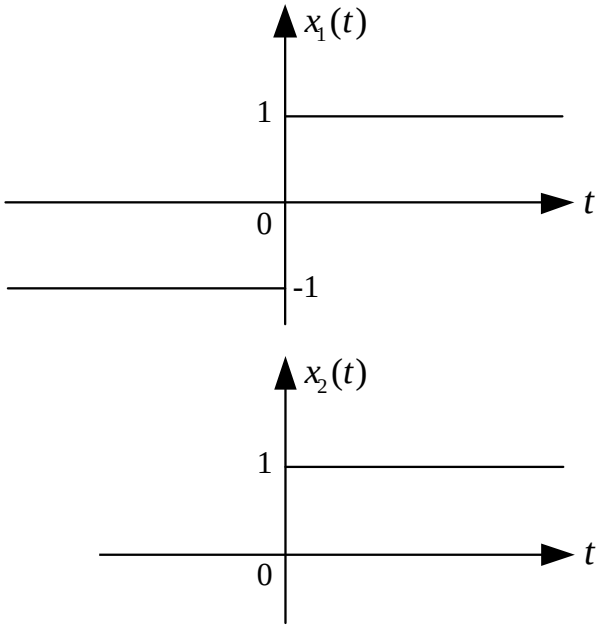


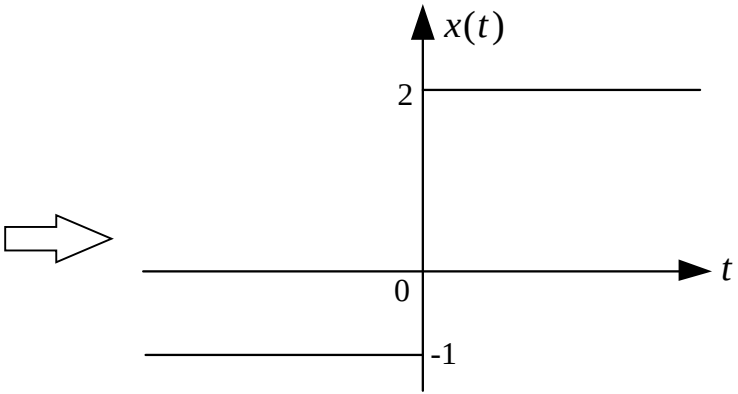
D



4. Addition

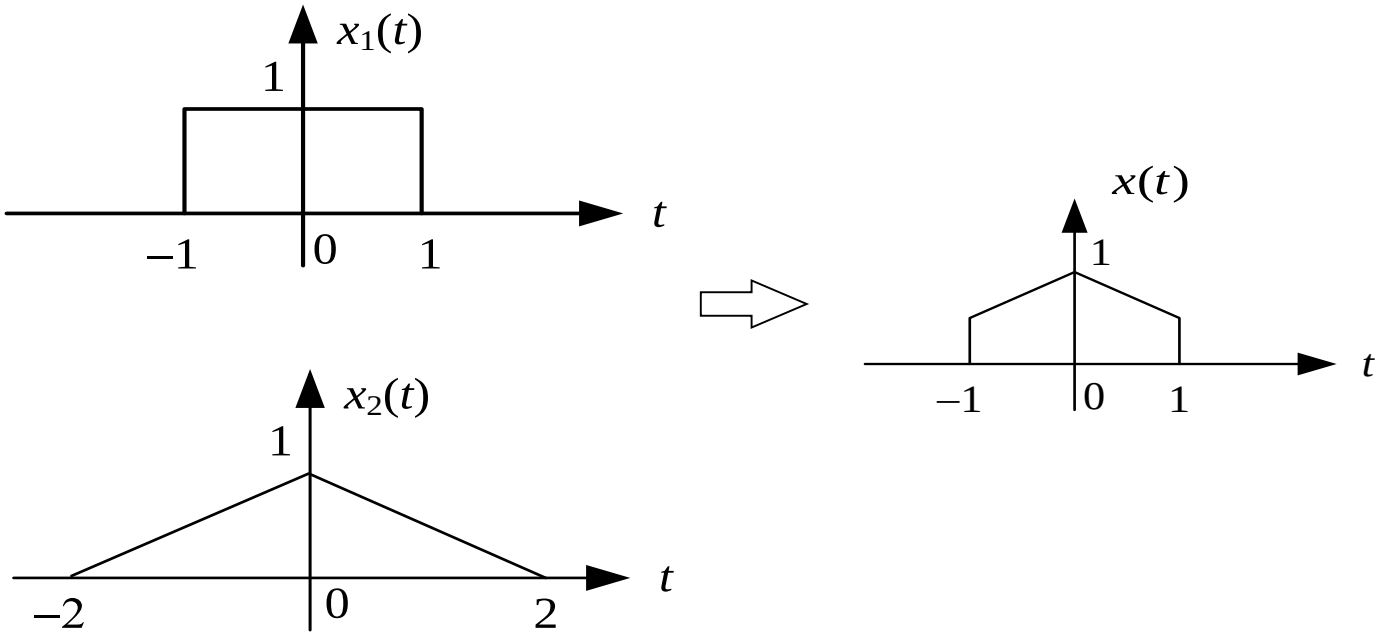
$$x(t) = x_1(t)+x_2(t)+ \ldots\ldots+x_n(t)$$



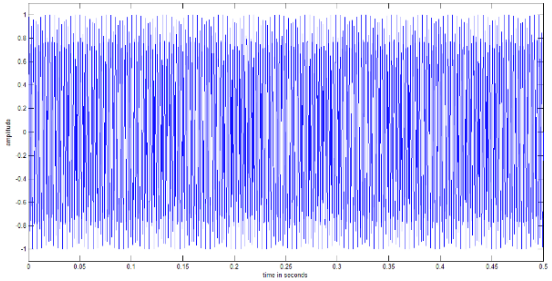


5. Multiplication

$$x(t) = x_1(t) \cdot x_2(t)$$

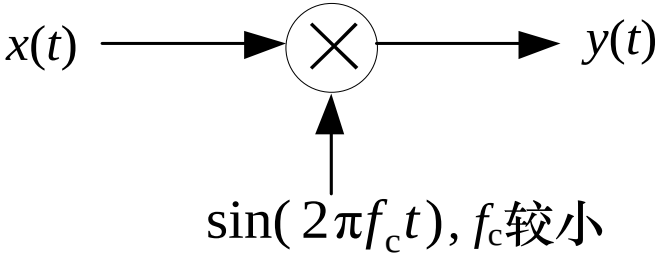


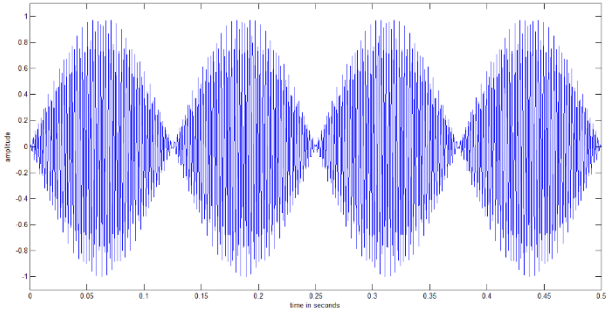
5. Multiplication



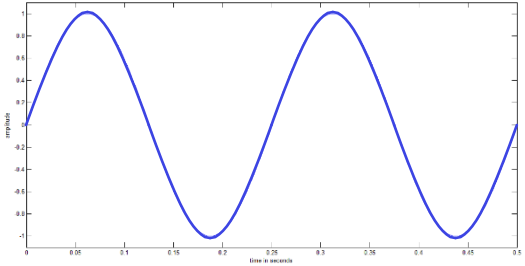
440Hz sine wave

Example: 440Hz sine multiplied by a 4Hz sine wave





After the tremor processing



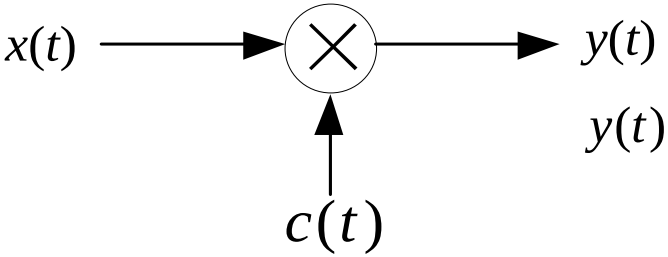
f_c=4Hz sine wave

5. Multiplication

Modulation is an important basis for various communication systems, and is widely used in radio, television, radar, measuring instruments and other electronic equipment.

Amplitude modulation

Modulate the signal

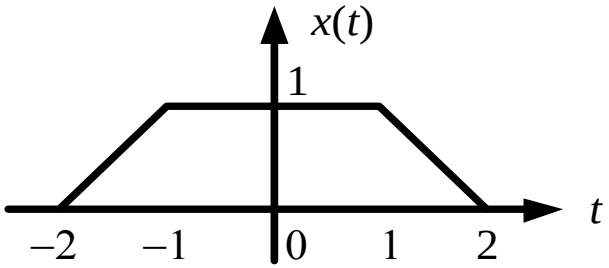


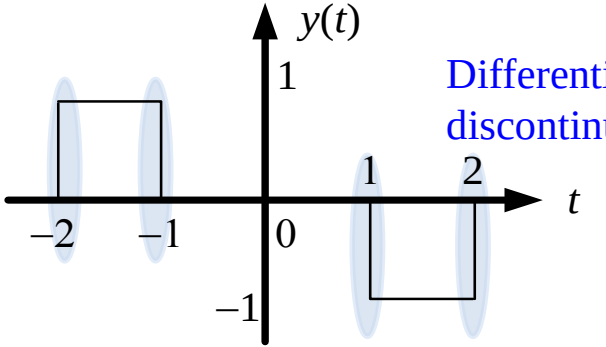
$$y(t) = x(t)c(t) = x(t)\cos(\omega_c t)$$

Carrier signal

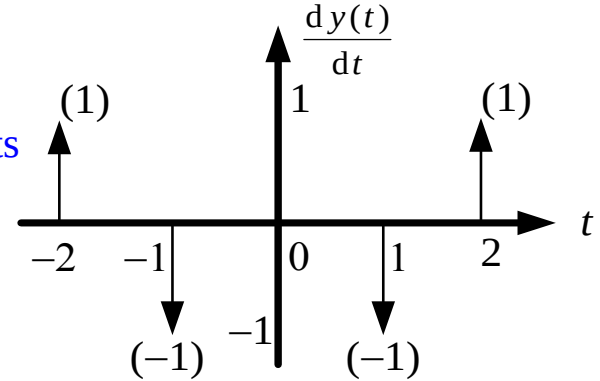
6. Signal differentiation

$$y(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} = x'(t)$$





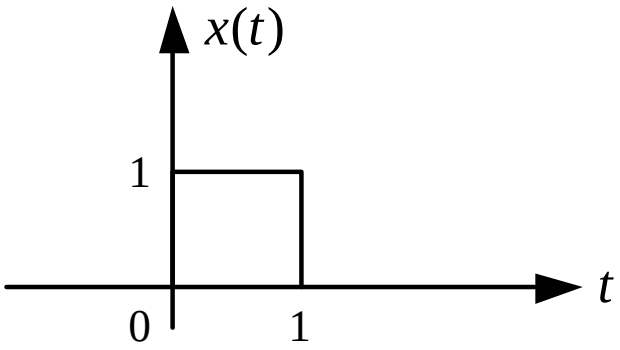
Differentiation of discontinuous points

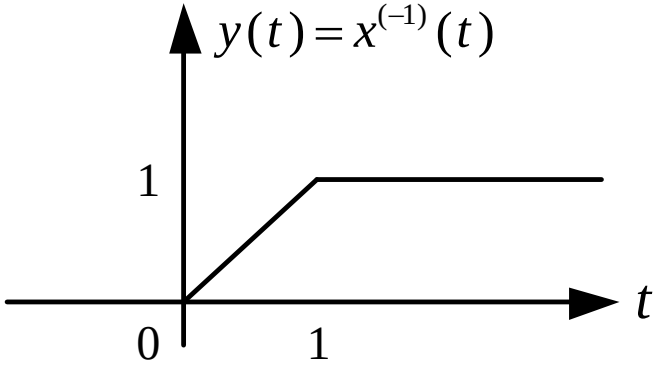


The impulse signal can represent the derivative of the signal at the signal break, and its strength value is the jump value of the signal break.

7. Signal Integration

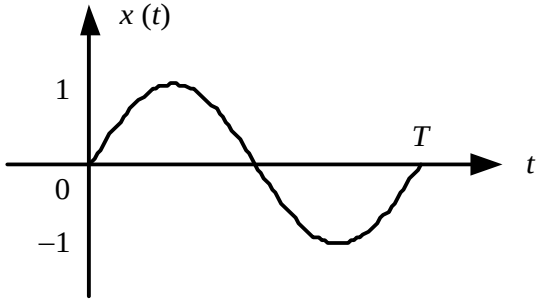
$$y(t)=\int_{-\infty}^{t} x(\tau)\cdot \mathrm{d}\tau = x^{(-1)}(t)$$





[Ex.] The x(t) waveform is known as follows, and the waveforms of its first derivative and second derivative are drawn, where T is constant, $\omega_0 = 2\pi/T$.

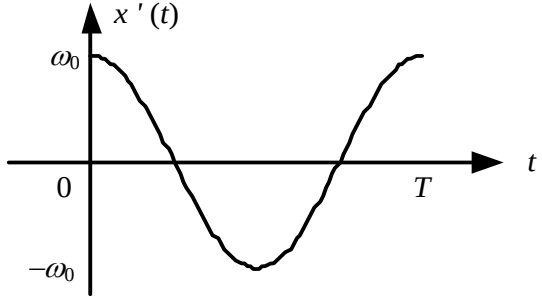
solution:



$$x(t) = \sin\omega_0 t \cdot [u(t) - u(t-T)]$$

$$x'(t) = \omega_0 \cos\omega_0 t \cdot [u(t) - u(t-T)] + \sin\omega_0 t \cdot [\delta(t) - \delta(t-T)]$$

$$= \omega_0 \cos\omega_0 t \cdot [u(t) - u(t-T)]$$

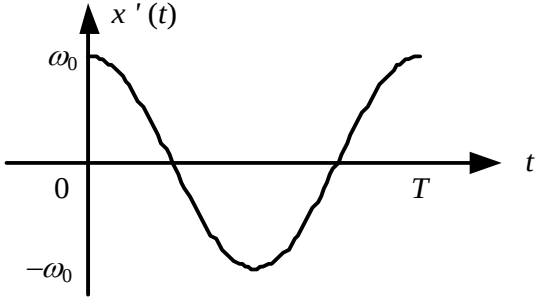


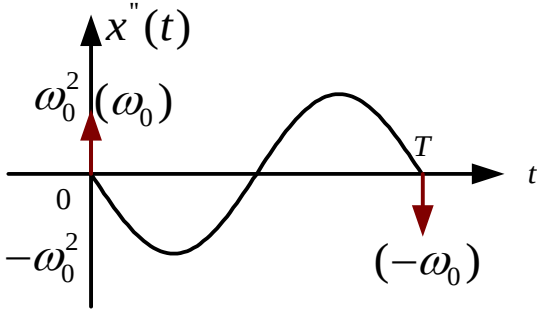
[Ex.] The x(t) waveform is known as follows, and the waveforms of its first derivative and second derivative are drawn, where T is constant, $\omega_0 = 2\pi/T$.

solution: $x'(t) = \omega_0 \cos\omega_0 t \cdot [u(t) - u(t-T)]$

$$x''(t) = \omega_0 \cos\omega_0 t \cdot [\delta(t) - \delta(t-T)] - \omega_0^2 \sin\omega_0 t \cdot [u(t) - u(t-T)]$$

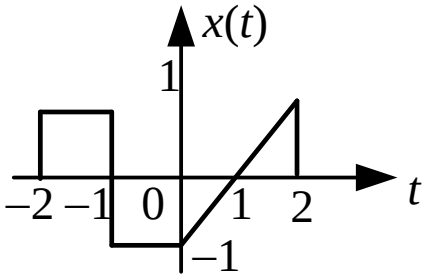
$$= \omega_0 \delta(t) - \omega_0 \delta(t-T) - \omega_0^2 \sin\omega_0 t \cdot [u(t) - u(t-T)]$$



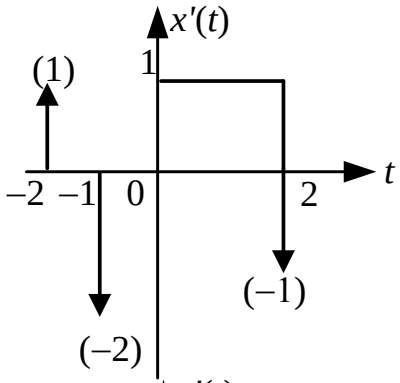


Single choice 1point

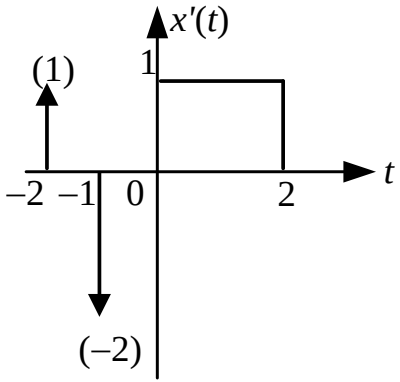
The waveform of the signal $x(t)$ is shown in the following figure, then the waveform of $x'(t)$ is



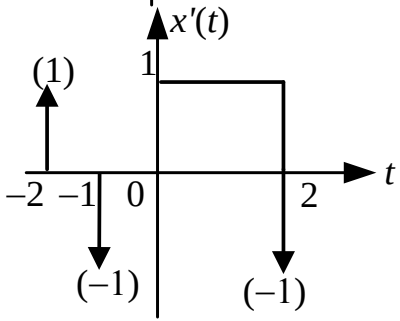
A



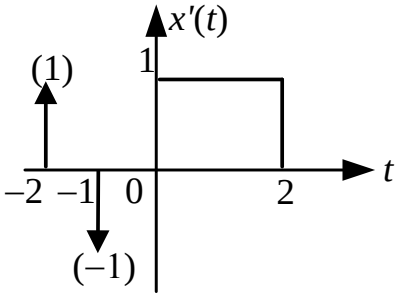
B



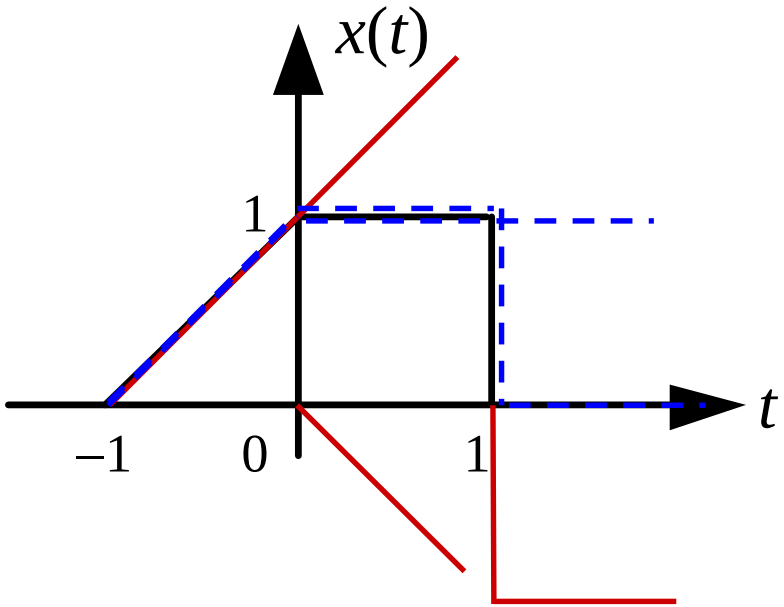
C



D

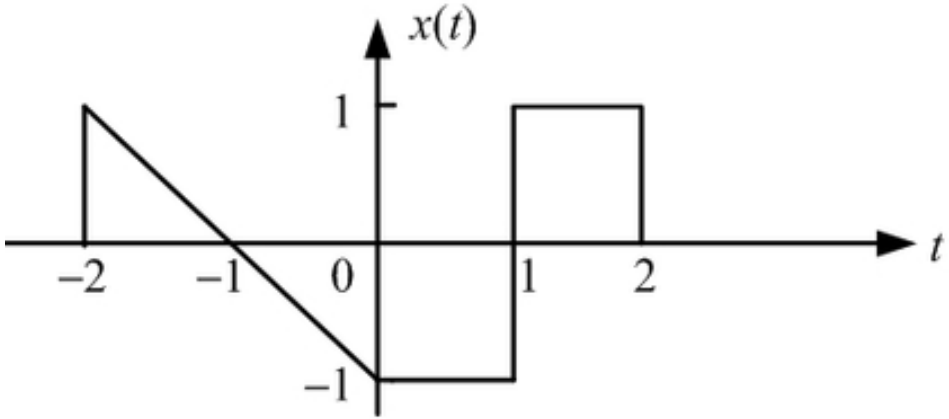


[Ex.] Try the basic signal to represent the signal shown in the figure $x(t)$



$$x(t) = r(t+1) - r(t) - u(t-1)$$

[Ex.] Try the basic signal to represent the signal shown in the figure $x(t)$



A. $u(t+2) - r(t+2) + r(t) + 2u(t-1)$

B. $u(t+2) - r(t+2) + r(t) + 2u(t-1) - u(t-2)$

C. $u(t+2) - 2r(t+2) + r(t) + 2u(t-1) - u(t-2)$

D. $r(t+2) + r(t) + 2u(t-1) - u(t-2)$

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!